

# Toward Personalized Hebrew ASR: Adapting a General Model to a Single Speaker

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## Abstract

Automatic speech recognition (ASR) systems are built for general-purpose use and evaluated by error rates averaged across many speakers, which says little about how well they serve any one person. Yet in many deployments it is a single speaker’s accuracy that matters. We ask how to personalize a Hebrew ASR model to one speaker under a small data and compute budget.

## 1 Introduction

Modern ASR models such as Whisper (Radford et al., 2023) are built for general-purpose use. Their quality is reported as an error rate averaged over many speakers and corpora, an average that can hide large differences: a model that serves most people well may still serve some of them poorly.

In many real deployments, however, the user is not the average speaker but one particular person: a Knesset member whose speeches are transcribed every day, a journalist dictating notes, or the single user of a voice assistant. For them, what matters is how well the model recognizes their own voice. This calls for personalization: adapting a general model to one speaker with a small amount of their own speech.

We study personalization for Hebrew and ask four questions: which fine-tuning strategy suits a single speaker, how much of their speech is needed, who benefits most, and whether a personal model outperforms one shared across speakers.

## 2 Related Work

**Averages hide who is served.** Whisper (Radford et al., 2023) and ivrit.ai’s Hebrew model (Marmor et al., 2025) are both evaluated by error rates averaged over large test sets, never per speaker or subgroup. Such averages have hidden gaps before: Whisper is much weaker on languages with little training data, Hebrew among them. ivrit.ai closed

this language gap by fine-tuning Whisper on about 5,000 hours of Hebrew, mostly Knesset plenary speech, producing ivrit-ai/whisper-large-v3, the starting point for our work.

**Gaps within a language.** Within a single language, too, ASR serves some people worse than others. Error rates differ by race (Koencke et al., 2020) and by gender, dialect, age and accent (Tatman, 2017; Feng et al., 2021; Liu et al., 2022). These studies compare groups, and we are not aware of such an analysis for Hebrew. We close this gap on the analysis side: we measure the error, and the gain from the Hebrew fine-tune, for each speaker and for subgroups of speakers who share attributes such as gender, age, origin or speaking rate.

**Adapting a model to one speaker.** Adapting a recognizer to its user is an old idea in speech recognition (Gales, 1998). With modern models, fine-tuning on a small amount of a person’s own speech can greatly reduce their errors, but this has mostly been studied for atypical speech, such as speech impairments and strong accents (Shor et al., 2019; Green et al., 2021). Training a full model for every speaker is expensive, so methods such as LoRA (Hu et al., 2022) train only a small number of added parameters, and have also been used with Whisper (Liu et al., 2024).

## 3 Experimental Setup

### 3.1 Overview

Our study consists of two experiments.

In the first experiment, we conduct a speaker-level performance analysis, and repeat it for subgroups of speakers who share attributes such as gender, age, origin or speaking rate. Both models transcribe the same held-out speech, allowing us to measure transcription error for each speaker and each subgroup. Because the second model is a He-

brew fine-tuned version of the first, we can also quantify the improvement attributable to Hebrew fine-tuning, per speaker and per subgroup.

In the second experiment, we focus on adaptation. From the speaker-level map, we select 12 speakers covering four profiles: speakers the Hebrew fine-tune helped least, speakers who remain hard under both models, typical speakers, and a low-WER speaker, plus two high-gain speakers as controls. To avoid selecting on measurement noise, speakers are ranked on their earlier sessions only, and their newest sessions are held out for testing. We then fine-tune the Hebrew model on each speaker’s own speech, and compare it with a control adapter trained on the same amount of speech pooled from other speakers. This separates learning the individual voice from learning committee speech in general.

### 3.2 Data

We use `ivrit-ai/knesset-committees`, a corpus of Israeli parliamentary committee sessions built from recorded audio–video and human-written protocols. The protocols are aligned to the recordings using forced alignment, providing transcript segments with speaker identities and alignment-confidence scores. Since the protocols are stenographic rather than fully verbatim, low-confidence segments may reflect mismatches between the text and audio.

The corpus spans November 2018 to January 2026 and contains approximately 13,177 sessions and 15,466 recorded hours, including about 7,886 hours of aligned speech with confidence scores above 0.7.

### 3.3 Models

We compare two models that share Whisper’s encoder–decoder architecture and differ in language-specific adaptation. Arm **A** is `openai/whisper-large-v3` (Radford et al., 2023), the general multilingual model, and arm **B** is `ivrit-ai/whisper-large-v3` (Marmor et al., 2025), the same model after fine-tuning on roughly 4,700 hours of Hebrew *plenary* speech. Arm **B** is also the starting point for personalization: all speaker and subgroup adaptation further fine-tunes `ivrit-ai/whisper-large-v3`. For the speaker-level analysis, arm **B** is served through `ivrit.ai`’s inference checkpoint (`whisper-large-v3-turbo-ct2`, with 4 decoder layers instead of 32).

### 3.4 Preprocessing

The corpus `ivrit-ai/knesset-committees` cannot be used directly in its released form. Whisper processes audio in segments of at most 30 seconds, whereas committee sessions can last several hours. In addition, our analysis is speaker-level, so each segment must be linked to a verified individual.

We therefore follow the preprocessing procedure introduced by VoxKnesset (Marmor et al., 2026), a 2,300-hour corpus of Hebrew parliamentary speech from 393 Knesset members with verified speaker identities and demographic metadata. We do not use VoxKnesset audio itself because it comes from the very plenary sessions arm **B** was fine-tuned on, so evaluating on it would test the model on speech it has already seen.

Speaker identification is performed first. Protocol speaker names are matched to the Knesset’s official roster of name variants and accepted only when consistent with the member’s term of service. Unresolved identities are discarded. During this stage, speaker-level attributes such as birth year, gender, and origin are also joined to the dataset using the matched Knesset member identity.

Each identified speaker turn is then divided into segments of at most 30 seconds without crossing speaker boundaries, with the aligned protocol text used as the reference transcript. Where dataset splits are required, they are session-disjoint within each speaker.

Each segment carries an alignment quality score between 0 and 1, indicating how well its protocol text is aligned to the audio (defined in Appendix A). Segments below 0.7 are excluded from all analyses. Training and validation data are further restricted to segments above 0.95, following `ivrit.ai`’s recommendation (Appendix A).

### 3.5 Speaker-Level Performance

To evaluate performance at the speaker level, we run inference with both models on the same set of audio segments: arm **A** through Hugging Face Inference Providers and arm **B** through a RunPod serverless endpoint hosting `ivrit.ai`’s inference checkpoint. To ensure comparability across speakers, we sample one hour of speech per speaker using a round-robin procedure over their sessions, so that the selected audio spans each speaker’s tenure.

For each speaker, we report WER and CER under both models, as well as the adaptation gain,  $WER_A - WER_B$ , and the relative gain,  $(WER_A -$

$WER_B)/WER_A$ . We also repeat the analysis across speaker subgroups defined by shared attributes.

### 3.6 Personalization

**Speakers and data.** We adapt the 12 selected speakers. To avoid leakage, each speaker’s meetings are split into training, validation and test sets that share no meeting, with the newest meetings held out for testing (Appendix D). Each speaker is trained on 5, 20 and 80 minutes of their own speech. The budgets are nested, filled from the speaker’s most recent training meetings first. Every model is scored on two test sets from the same held-out meetings (Appendix A): the high-quality segments (*clean test*, median 47 minutes per speaker) and all segments at quality  $\geq 0.7$  (*full test*, median 90 minutes).

**Adapter and training.** We train LoRA adapters (Hu et al., 2022) of rank 8 on the query and value projections of every attention layer of arm B (3.9M trainable parameters), for at most 400 steps with early stopping: validation loss is measured every 20 steps on other meetings of the speaker, used for neither training nor testing, and training stops after four checks without improvement, restoring the best checkpoint. The training labels are the protocol text.

**Control.** An adapter trained on a speaker’s audio learns both the speaker and the committee setting: rooms, microphones, vocabulary and the protocol’s style. To separate the two, the 12 speakers are split at random into two folds of six. For each fold and budget, a *control* adapter is trained with the same recipe on the same number of minutes from the six speakers of the *other* fold. Each speaker is scored on the same test segments under their own adapter and under the control, which never heard them. We report the relative gain over arm B,  $(WER_B - WER_{\text{adapted}})/WER_B$ , and the **personalization**,  $(WER_{\text{control}} - WER_{\text{own}})/WER_B$ , which equals the own gain minus the control gain. Its 95% interval and p-value come from a paired bootstrap over test segments (Appendix E). All budgets use seed 0; at 80 minutes, seeds 1 and 2 repeat the own adapters and the controls, with the folds redrawn, to distinguish speaker-level effects from variance caused by the random seed.

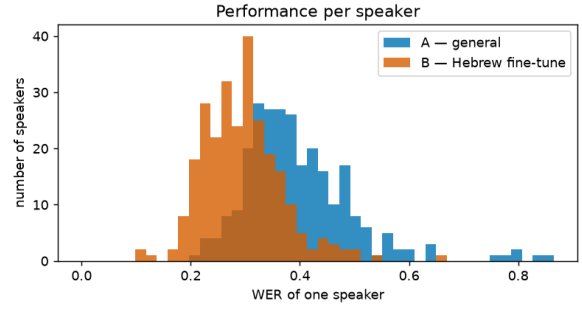


Figure 1: Per-speaker WER under arm A (general) and arm B (Hebrew fine-tune), all 267 speakers. Arm B shifts the whole distribution down without narrowing it in proportion.

## 4 Results

### 4.1 Speaker-Level Error Map

After the quality filter, the analysis covers 267 speakers and 217 hours (58,180 segments), with a corpus WER of 0.387 for arm A and 0.292 for arm B. Statistics below are over the 261 speakers with at least 20 segments, and per-speaker 95% confidence intervals come from bootstrapping each speaker’s segments (Appendix F).

**The fine-tune helps every speaker, but not equally.** All 261 speakers have a lower WER under arm B, and for every one of them the improvement is beyond sampling noise: the 95% bootstrap interval of their gain lies entirely above zero (Figure 9, Appendix D). Yet the fine-tune does not make speakers more equal (Figure 1): the spread of per-speaker WER shrinks (standard deviation 0.099 to 0.069) only in proportion to the lower average, so relative to the mean, speakers differ as much as before (coefficient of variation is 0.25 vs. 0.23). Speakers keep their order (Spearman  $\rho = 0.90$  between  $WER_A$  and  $WER_B$ ), and this ranking is stable across a speaker’s recordings (Appendix D).

**Group membership explains little.** We compare the gain across seven groupings of speakers: gender, nationality, religion, religious orientation, country of origin, age quartile and speaking-rate tertile. The clearest difference is speaking rate (Figure 2, with plots for the other six groupings in Figure 11, Appendix D): slow speakers gain more than fast ones (27% vs. 22% of arm A’s error removed). Arab, Muslim, Druze and Bedouin speakers also gain more than Jewish speakers (29–30% vs. 24%). However, group sizes are highly imbalanced (e.g., 229 Jewish speakers against 5

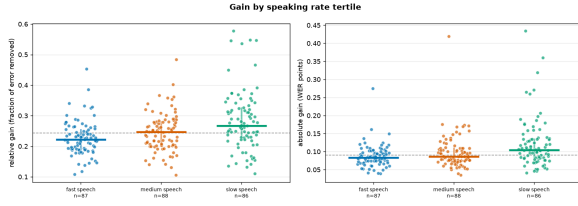


Figure 2: Gain from the Hebrew fine-tune by speaking-rate tertile, relative (left) and absolute (right). Each point is one speaker, the bar is the group median, and the dashed line is the overall median.

Bedouin), and the variation within each group far exceeds the differences between group medians. The bottom line: a speaker’s group says little about how well the model serves them, which motivates adapting to individual speakers. We test shared adaptation only on a randomly pooled group of speakers, and leave further analysis and fine-tuning on demographically defined subgroups to future work.

**Speaker selection.** From this map we select 12 speakers for personalization, covering five profiles: speakers the Hebrew fine-tune helped least, speakers who remain hard under both models, typical speakers, a low-WER speaker to test how much headroom is left, and two high-gain controls.

## 4.2 Adaptation Results

**The 80-minute adapter improves almost every speaker.** In Figure 3 (blue, 80 minutes), the own adapter’s gain (its relative WER reduction from arm B) with seed 0 has a median over the 12 speakers of +12.0% on the clean test and +15.3% on the full test; it is above zero for 11 of 12 speakers on the clean test (9 significantly) and for all 12 on the full test (all significantly). The other two seeds agree: over all 36 speaker–seed pairs, the adapter beats arm B in 35 on the clean test and in all 36 on the full test, and with each speaker’s gain averaged over the three seeds (Figure 4, column “own 80 min”), the clean-test gains have a median of +11.4% and a mean of +13.2%, ranging from +3% to +28%.

**How Much Speaker Data Is Needed.** The own adapter’s median relative gain over arm B grows from +4.3% to +7.5% to +12.0% on the clean test at 5, 20 and 80 minutes, and from +4.2% to +9.7% to +15.3% on the full test (Figure 3); no 5- or 20-minute adapter is the best model for any speaker (Figure 4). The control rises almost as fast, to

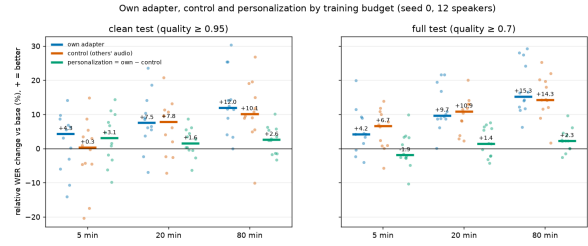


Figure 3: Relative WER change over arm B at 5, 20 and 80 minutes of training audio (seed 0, 12 speakers), on the clean test (quality  $\geq 0.95$ ) and the full test (quality  $\geq 0.7$ ). Blue: the speaker’s own adapter; orange: the control, trained on the same minutes from other speakers; green: personalization (own – control). Each dot is one speaker; bars and numbers are medians. The median personalization is the median of per-speaker differences, not the difference of the two medians.

+0.3%, +7.8% and +10.1% on the clean test and +6.7%, +10.9% and +14.3% on the full test, so what extra audio buys is mostly not specific to the speaker, and the personal margin stays between +1.6% and +3.1% on the clean test at every budget. At 5 minutes both adapters are unreliable: they are worse than arm B for some speakers, and the personal margin has opposite signs on the two test sets.

**Personal versus Shared Adaptation.** At 80 minutes, the median personalization over the 36 speaker–seed pairs is +2.4% on the clean test and +2.2% on the full test. With seed 0, the own adapter gains +12.0% and +15.3%, while the control, which never heard the speaker, already gains +10.1% and +14.3% (Figure 3): most of the adaptation gain is general, and the speaker’s own audio adds 2–3% on top of it.

Five speakers are significant in all three seeds on the full test (Figure 5), led by 23558 at +9% to +10%; six never are, in any seed or test. Profile does not predict the effect: speakers the fine-tune helped least and speakers hard under both models fall on both sides. A speaker’s personalization varies across seeds with a median standard deviation of 2.4% on the clean test, as large as the effect, so we make per-speaker claims only where all seeds agree. Nor does the margin grow on cleaner audio: by alignment-quality band of the full test, own adapter and control gain alike on the two hardest bands (about +20%), and the median personalization is +0.5%, +4.0%, +2.5% and +2.3% from the hardest band to the cleanest.

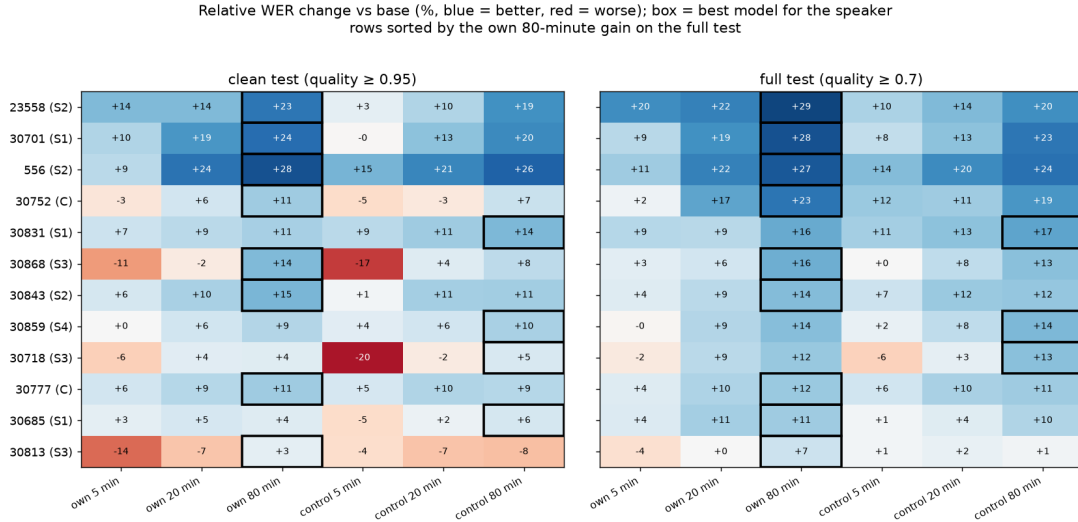


Figure 4: Relative WER change over arm B (%) for each speaker under each adapter; blue is better, red is worse, and the box marks the speaker’s lowest WER. The 80-minute columns are averaged over three seeds. Rows are sorted by the own 80-minute gain on the full test.

**Learning Curves.** Validation loss reaches its minimum at step 60–100, about two passes over the data and 38–54% below the validation loss of arm B without an adapter, and then rises as the adapter overfits (Figure 6). Every run stops early, by step 180 of 400, and the three seeds trace nearly identical curves (Figure 15, Appendix D). An adapter trains in under two minutes on one GPU.

## 5 Conclusion

The Hebrew fine-tune lowers every speaker’s error but leaves them as unequal as before, which motivates adapting to individual speakers.

**Most of the adaptation gain is general.** An 80-minute LoRA adapter on a speaker’s own committee speech lowers arm B’s WER by a median 12% on the clean test and 15% on the full test; the same amount of other speakers’ audio already lowers it by 10% and 14%. Training on the speaker’s own audio adds a further 2–3% over this shared baseline, an effect that is stable across seeds.

**More audio gives more gain, but mostly not personal gain.** From 5 to 80 minutes the own adapter’s gain roughly triples, and the control’s grows almost as fast.

**Adaptation is cheap.** An adapter converges in about two passes over its data. The recipe is insensitive to rank, augmentation and, at 80 minutes, the learning rate, but Whisper’s dropout prevents learning.

Whether the personal margin grows with substantially more speaker data is tested in a follow-up on four speakers, now running. Further directions are adaptation on similar rather than random speakers and evaluation against verbatim human transcripts.

## Limitations

**Prior exposure.** Arm B was fine-tuned on Kneset plenary speech, which includes many of our speakers. Our evaluation uses committee recordings, a different setting and register, but the model may still be partly familiar with these voices.

**Shared recording conditions.** Train and test sets are meeting-disjoint, but not committee-disjoint: about 75–82% of the test speech comes from committees the same speaker also appears in during training, and so likely from the same rooms and microphones. Part of an adapter’s gain may therefore reflect familiar recording conditions rather than the voice alone.

**The reference is the protocol.** Every WER is measured against the stenographic protocol, which is not a verbatim transcript. Our gains are gains in agreement with the protocol, and may partly reflect learning what the stenographer omits. We limit this by training and validating only on segments whose protocol closely matches the audio (quality  $\geq 0.95$ , Appendix A). That score, however, comes from a Whisper-family aligner, so a test set filtered the same way would favour speech that Whisper models already transcribe well; we therefore score

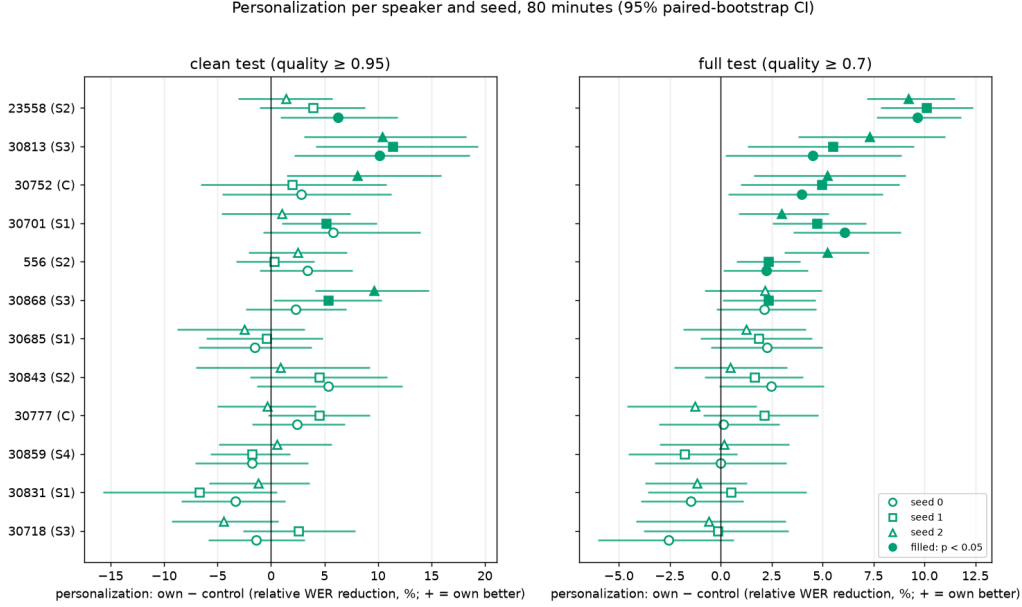


Figure 5: Personalization (own – control, relative WER reduction in %) per speaker and seed at 80 minutes, on the clean test (left) and the full test (right), with 95% paired-bootstrap intervals. Filled markers: own adapter significantly better than the control (paired bootstrap,  $p < 0.05$ ); hollow: not significant. Speakers are sorted by their mean on the full test. Profiles: S1, helped least by the fine-tune; S2, hard under both models; S3, typical; S4, low WER; C, high-gain control.

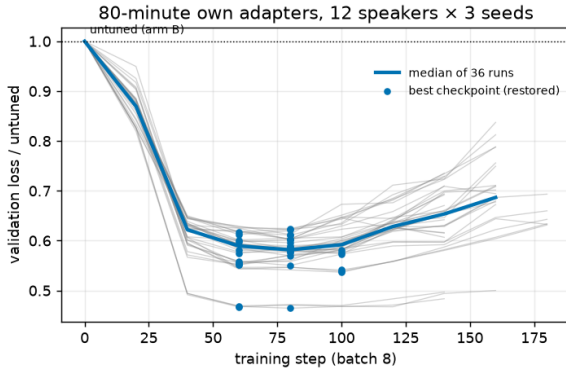


Figure 6: Validation loss relative to arm B while training the 80-minute own adapters (12 speakers  $\times$  3 seeds). Thin lines: one run each; thick: the median; dots: the checkpoints early stopping restored.

every model both on the high-quality segments and on all segments at quality  $\geq 0.7$ . Whether the personal margin is larger against what was actually said requires a human verbatim reference.

**Reference transcripts.** Our references are the Knesset’s stenographic protocols, not verbatim transcripts: stenographers condense, rephrase and omit speech, so part of every measured error is a mismatch between protocol and audio rather than a recognition mistake. We limit this with alignment-quality filters set in consultation with the ivrit.ai

team (Section 3.4, Appendix A), but no verbatim, human-transcribed test set was available. Absolute WERs therefore partly reflect the protocol’s editing, and our comparisons are most reliable between models scored against the same references.

**Limited compute.** Training and inference ran on paid cloud services (RunPod GPUs and Hugging Face inference), at a total cost of about \$200 across all our experiments. Within this budget we focused on the experiments central to our question, and a larger budget would allow more models, configurations and speakers to be explored.

**Tuning on validation loss.** The rank, dropout, augmentation and learning rate were chosen on the validation loss of four speakers. The rejected settings, including rank 16, were never scored on the test sets.

**A random control group.** The control pools speakers at random, so it does not test whether *similar* speakers could replace a speaker’s own audio; we leave that comparison to future work.

## Acknowledgments

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## A Data Preparation Details

**Alignment quality score.** The corpus aligns each protocol to its recording with a Whisper-family forced aligner, which assigns every protocol word a time span and a probability: how likely the model finds that word at that point in the audio. A protocol segment’s quality score is the median probability of its words. Our segments of up to 30 seconds can join several protocol segments, so their score is the duration-weighted median over those parts. A low score means the text and the audio disagree, potentially because the stenographer condensed or omitted speech, or because of cross-talk. ivrit.ai treat 0.65 as fit for training and 0.95 as high quality.

**Quality filter for the analysis.** The 0.7 threshold removes 7,810 of 65,990 segments (11.8%, 12.8 hours). Removing them lowers the corpus WER from 0.417 to 0.387 (A) and from 0.324 to 0.292 (B), and leaves the difference between the arms almost unchanged.

**Quality filter for training.** Because the score is a median, a segment can pass it while a few of its words are misaligned. Training and validation segments therefore use the word probabilities directly and must pass four conditions: confidence at least 0.95, at most 10% of words with probability below 0.5, no run of three or more such words, and the number of aligned words falling within the segment’s time span matches the number of words in its reference text to within 20%, so that the word probabilities checked above actually cover the segment’s text.

**Two test sets.** The aligner is itself a Whisper-family model, so a 0.95 threshold favours segments that Whisper models already transcribe well. Applied to the test set, this introduces a selection bias toward easy speech. We therefore score every model on two test sets drawn from the same held-out sessions: the filtered segments, and all segments with confidence of at least 0.7.

## B Training Details

All settings were chosen on validation loss alone, in 64 tuning runs that never touched the test sets; the result is Table 1, shared by every own and control adapter.

**Tuning procedure.** Four tuning speakers, one per profile (30685, S1; 23558, S2; 30718, S3; 30859, S4), were trained at the two ends of the budget range, 5 and 80 minutes; 20 minutes takes the 80-minute choice. The criterion is the relative drop in validation loss from arm B to the best checkpoint. Tuning ran in three stages, each starting from the previous stage’s choice: the learning rate ( $10^{-4}$ ,  $3 \times 10^{-4}$ ,  $10^{-3}$ ); LoRA rank  $\{8, 16\} \times$  Whisper’s dropout  $\{0, 0.1\}$ ; and augmentation (SpecAugment, alone or with  $\pm 10\%$  tempo changes; no pitch or speed perturbation, which would change the voice). A setting replaces the simpler incumbent ( $3 \times 10^{-4}$ , rank 8, no dropout, no augmentation) only if it raises the mean drop by at least 0.01 and helps at least three of the four speakers.

**Tuning results** (Figures 7 and 8).

- **Learning rate matters only at 5 minutes.** There,  $10^{-4}$  gives a mean drop of 0.294 against 0.266 and 0.269 for the higher rates, whose best checkpoint comes at step 20–40, after which their validation loss rises above arm B’s. At 80 minutes the three rates give 0.434, 0.432 and 0.429: early stopping absorbs the difference, so the incumbent  $3 \times 10^{-4}$  stays.
- **Rank 16 adds nothing:** 0.432 at 80 minutes, equal to rank 8, and 0.281 against 0.294 at 5 minutes.
- **Whisper’s dropout prevents learning.** At 0.1, the drop collapses to 0.019 at 5 minutes and 0.077 at 80 for every speaker.
- **Augmentation is within noise:** 0.298 and 0.300 at 5 minutes (below the 0.01 threshold), 0.430 and 0.431 at 80.

## C Implementation Details

**Training.** Adapters are trained with PyTorch, Hugging Face transformers (Seq2SeqTrainer) and peft, on the frozen base model in bfloat16. Training segments are fed latest meeting first, so the 5-, 20- and 80-minute budgets are nested prefixes of one ordering. Each control’s pool takes one segment from each of its six speakers in turn until the budget is reached.

| Hyperparameter                 | Value                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------------|
| Base model                     | ivrit-ai/whisper-large-v3, frozen, bfloat16                                                                   |
| Adapter                        | LoRA on the query and value projections of every attention layer (encoder, decoder self- and cross-attention) |
| Rank / $\alpha$ / LoRA dropout | 8 / 16 / 0.05                                                                                                 |
| Trainable parameters           | 3,932,160                                                                                                     |
| Learning rate                  | $10^{-4}$ at 5 minutes; $3 \times 10^{-4}$ at 20 and 80 minutes                                               |
| Schedule                       | constant after 20 warm-up steps                                                                               |
| Batch size                     | 8, no gradient accumulation                                                                                   |
| Weight decay                   | 0.01                                                                                                          |
| Whisper dropout                | 0                                                                                                             |
| Augmentation                   | none                                                                                                          |
| Training length                | at most 400 steps                                                                                             |
| Validation                     | loss every 20 steps                                                                                           |
| Early stopping                 | after 4 validations without improvement; best checkpoint restored                                             |
| Training labels                | protocol text                                                                                                 |
| Seeds                          | 0 at every budget; 1 and 2 at 80 minutes                                                                      |

Table 1: Training hyperparameters, identical for every own and control adapter.

**Scoring.** Reference and hypothesis are normalized (niqqud and punctuation removed) and aligned word by word; substitutions, deletions and insertions are counted per segment. WER is pooled over a speaker’s test set: the sum of word errors divided by the sum of reference words. Confidence intervals and p-values come from a paired bootstrap over test segments (2,000 resamples, two-sided p); for personalization, the own and control hypotheses of the same segments are resampled together.

## D Additional Results

### D.1 Speaker-Level Error Map

**Reliability of the per-speaker measures.** Each speaker is measured on about one hour of speech, so their WER and gain carry sampling noise. To test whether the differences between speakers are a stable property of the speakers rather than of the particular sample, we split each speaker’s sessions into two alternating halves, compute  $WER_B$  and the relative gain on each half independently, and compare the halves across speakers (Figure 10). If the differences were noise, the two halves would be unrelated and the points would form a shapeless cloud. If they are stable, the points follow the diagonal. They do: the correlation between halves, corrected to full length (Spearman–Brown), gives a reliability of 0.78 for  $WER_B$  and 0.75 for relative

Tuning runs: every setting tried, per speaker (rank 8, dropout 0, no augmentation unless stated)

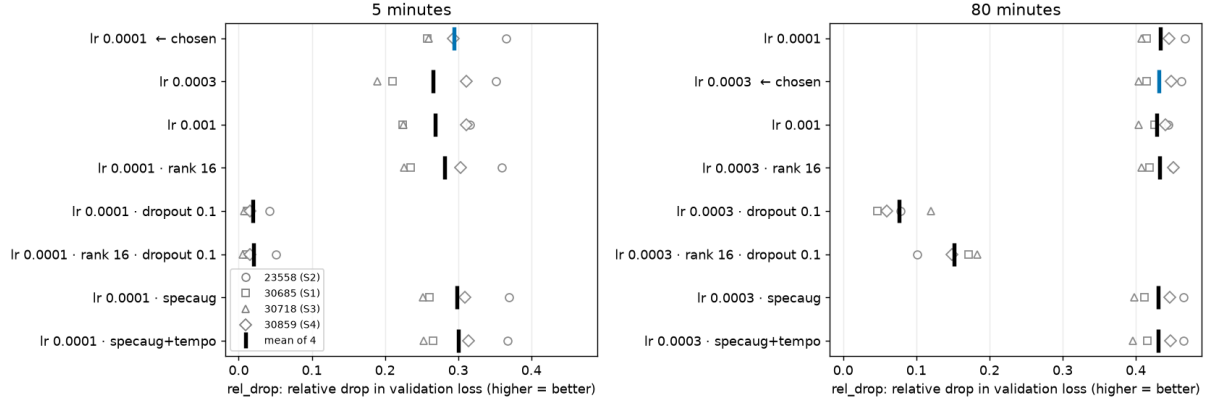


Figure 7: Every tuning run, at 5 minutes (left) and 80 minutes (right). Markers: the four tuning speakers; bar: their mean; blue: the chosen setting. The x-axis is the relative drop in validation loss from arm B to the best checkpoint.

Tuning curves on validation (4 speakers; thin = speaker, thick = mean, dot = best checkpoint)

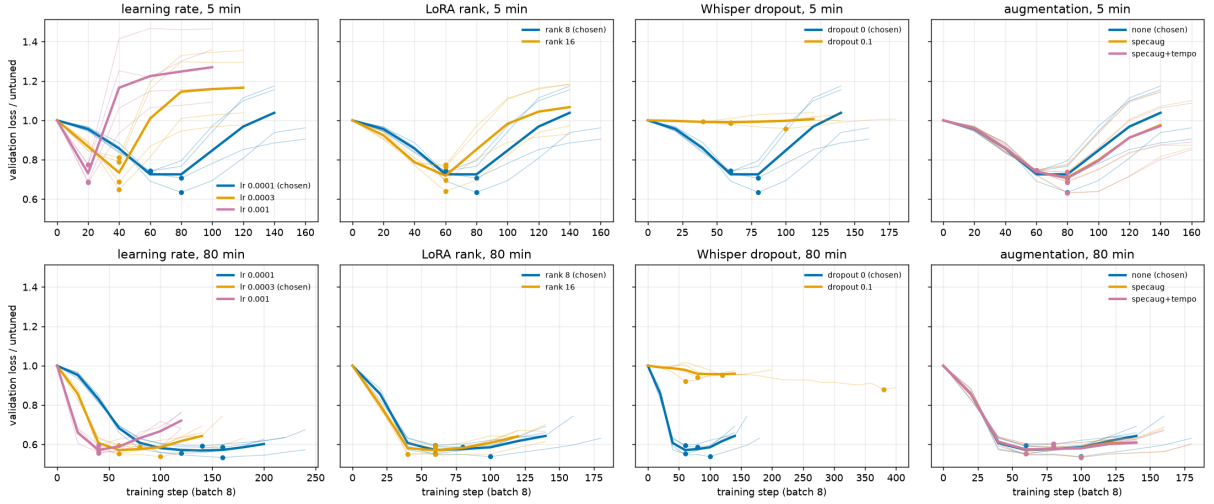


Figure 8: Validation loss relative to arm B during tuning, varying one setting at a time around the chosen recipe, at 5 minutes (top) and 80 minutes (bottom). Thin lines: the four speakers; thick: their mean; dots: the best checkpoints.

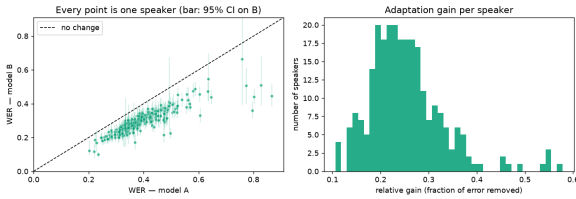


Figure 9: Left: per-speaker WER under arm A against arm B. Every point is one speaker, the bar is the 95% bootstrap interval on  $WER_B$  and the dashed line is no change. Right: distribution of the relative gain over the 261 speakers with at least 20 segments.

gain. This is what justifies ranking speakers by their gain when selecting them for personalization.

**Gain by group.** Figure 11 shows the gain for the six groupings not shown in the main text (speaking

rate is Figure 2). Speakers with missing metadata and groups of fewer than three speakers are omitted.

**Protocol omissions.** Over half of arm B’s errors are insertions (Table 2), and 73% of its inserted words also appear in arm A’s transcript of the same segment. Agreement between the two models does not show that a word was spoken: on a sample of human-corrected clips, only about half of such words (19 of 36) were. Part of arm B’s insertion errors therefore reflects speech the stenographic protocol left out, and part reflects genuine recognition errors. All WERs in this paper are computed against the protocol as is.

**Recording conditions.** WER also varies with the recording conditions: arm B scores 0.32–0.37

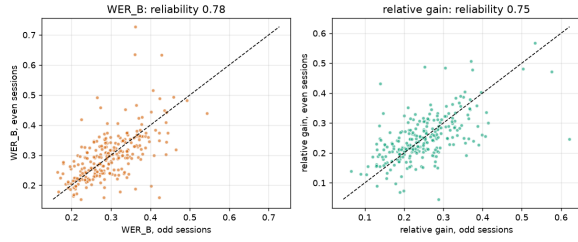


Figure 10: Each point is one speaker, measured separately on their odd and even sessions. Points near the diagonal mean the measure repeats across recordings, and titles give the split-half reliability (Spearman–Brown corrected).

| Model                | WER   | CER   | share of word errors |      |      |
|----------------------|-------|-------|----------------------|------|------|
|                      |       |       | Sub.                 | Del. | Ins. |
| A (general)          | 0.387 | 0.241 | 49%                  | 13%  | 38%  |
| B (Hebrew fine-tune) | 0.292 | 0.203 | 36%                  | 11%  | 53%  |

Table 2: Corpus error rates after the quality filter (267 speakers, 58,180 segments) and the composition of each model’s word errors: substitutions, deletions and insertions.

in 2018–2019 and 0.27–0.29 from 2020 on (Figure 12), and 0.21–0.37 across the twelve largest committees. Personalization splits are therefore meeting-disjoint and ordered by date: each speaker is adapted on earlier meetings and tested on their newest ones, which is also how a personalized model would be used in practice.

## D.2 Personalization

### E Significance Testing

Every comparison in Section 4.2 is paired: two models transcribe the same test segments of one speaker, so we ask whether one makes fewer errors than the other on those segments.

**What is compared.** For the gain of an adapter over arm B, the difference is  $WER_B - WER_{\text{adapted}}$ . For personalization, it is  $WER_{\text{control}} - WER_{\text{own}}$ : the own adapter and the control on the same segments. Both are pooled differences, the sum of each model’s word errors over the segments divided by the sum of reference words.

**Resampling.** We draw as many segments as the speaker’s test set holds, with replacement, and recompute the pooled difference on the draw; this is repeated 2,000 times. The same draw is applied to both models, so a hard segment counts for both or for neither, and the variation between

segments largely cancels. We resample whole segments rather than words because errors within a segment are not independent: a misheard name or a runaway transcription produces several at once.

**Interval and p-value.** The 95% interval is the 2.5th to 97.5th percentile of the 2,000 differences, in absolute WER; dividing by  $WER_B$  gives it in the relative units of Section 4.2. The two-sided p-value is twice the smaller of the shares of draws at or below zero and at or above zero, so it resolves p-values down to 0.001; smaller ones are reported as  $p < 0.001$ .

**Multiple comparisons.** Each speaker–seed pair is tested on its own, without correction. If own adapters and controls did not differ, about two of 36 pairs would fall below  $p = 0.05$  by chance, and about one of them with a positive sign; we observe 8 (clean test) and 16 (full test) significantly positive pairs and none significantly negative. Per-speaker claims rest on agreement across all three seeds rather than on a single p-value.

## F Bootstrap for the Speaker-Level Error Map

The per-speaker intervals of Section 4.1, and the bars in Figure 9, come from a simpler bootstrap than Appendix E’s, run once per speaker on arm A’s and arm B’s transcriptions of that speaker’s one-hour sample.

**Resampling.** The speaker’s segments are drawn with replacement, as many as the sample holds, 1,000 times. The same draw is used for both arms. Each draw gives a pooled  $WER_A$  and  $WER_B$  (the sum of word errors over the drawn segments divided by the sum of reference words) and their difference, the absolute gain  $WER_A - WER_B$ .

**Intervals.** The 95% interval of  $WER_A$ ,  $WER_B$  and the gain is the 2.5th to 97.5th percentile of the 1,000 values. No p-value is computed: a speaker’s improvement under arm B is called beyond sampling noise when the interval of their gain lies entirely above zero, as it does for all 261 speakers with at least 20 segments.

**Difference from Appendix E.** The resampling is the same (whole segments, paired across the two models), but it uses 1,000 draws instead of 2,000, compares the two base models instead of adapters, and reports intervals only.

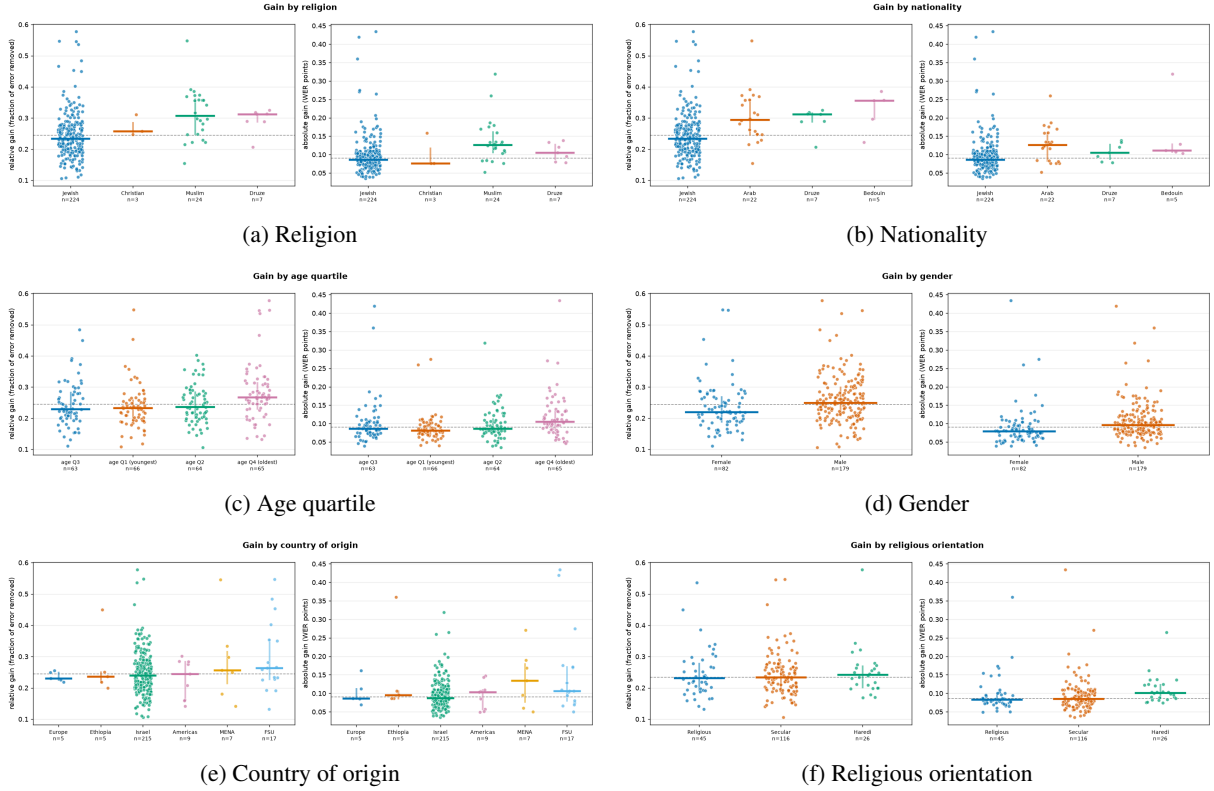


Figure 11: Gain from the Hebrew fine-tune by group, relative (left of each panel) and absolute (right). Each point is one speaker with at least 20 segments. The bar is the group median, the line the interquartile range, and the dashed line the overall median.

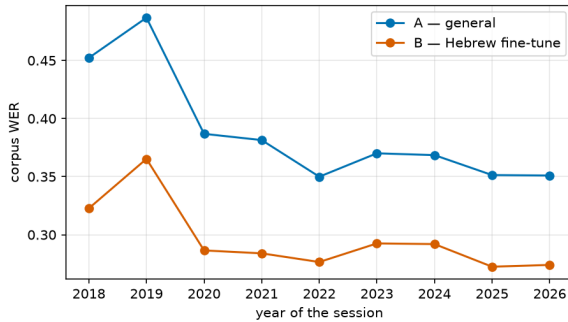


Figure 12: Corpus WER of both models by the year of the session, after the quality filter.

## G Division of Work

Both authors shaped the project from the start. We developed the idea together over several brainstorming meetings, refining a general direction into the final research question. We then sharpened it in discussions with the ivrit.ai team, who pointed us to directions and problems they had encountered and considered worth studying.

We set the guidelines of each stage together and divided the detailed work between us. For the speaker-level analysis (Stage 1), we jointly defined

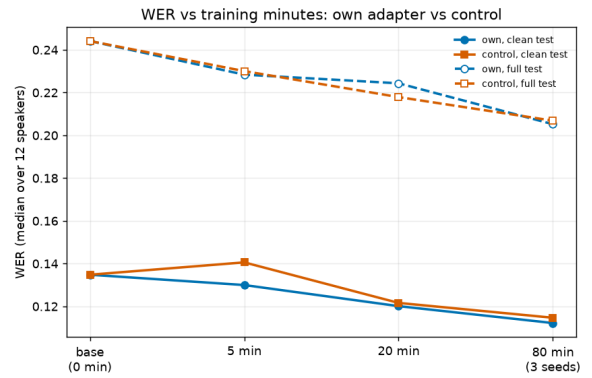


Figure 13: Median WER over the 12 speakers at 0 (arm B), 5, 20 and 80 minutes of training audio, for the own adapter (blue) and the control (orange), on the clean test (solid) and the full test (dashed). At 80 minutes, each speaker's WER is averaged over three seeds.

the speaker groupings and the analyses to run, and Hadas Yonat, a data engineer, carried out most of the in-depth analysis. For personalization (Stage 2), we jointly designed the main principles of the training architecture, and Dolev Abudi led the training itself. Throughout, we worked in parallel and in close coordination: in regular joint working sessions, in person and over Zoom, we shared our

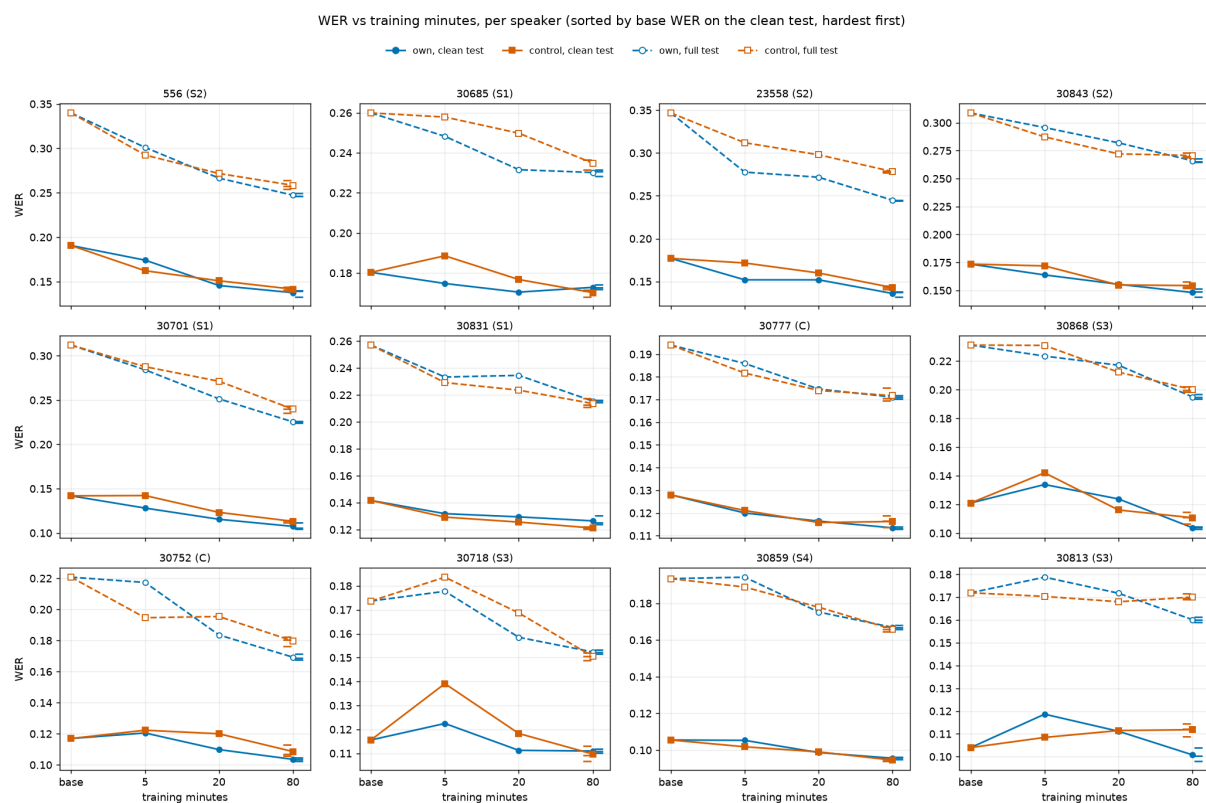


Figure 14: The same per speaker, sorted by arm B's WER on the clean test, hardest first. Ticks at 80 minutes: the individual seeds.

progress and findings and made the project's decisions together. We enjoyed working together, and this teamwork was an important and meaningful part of the project for both of us.

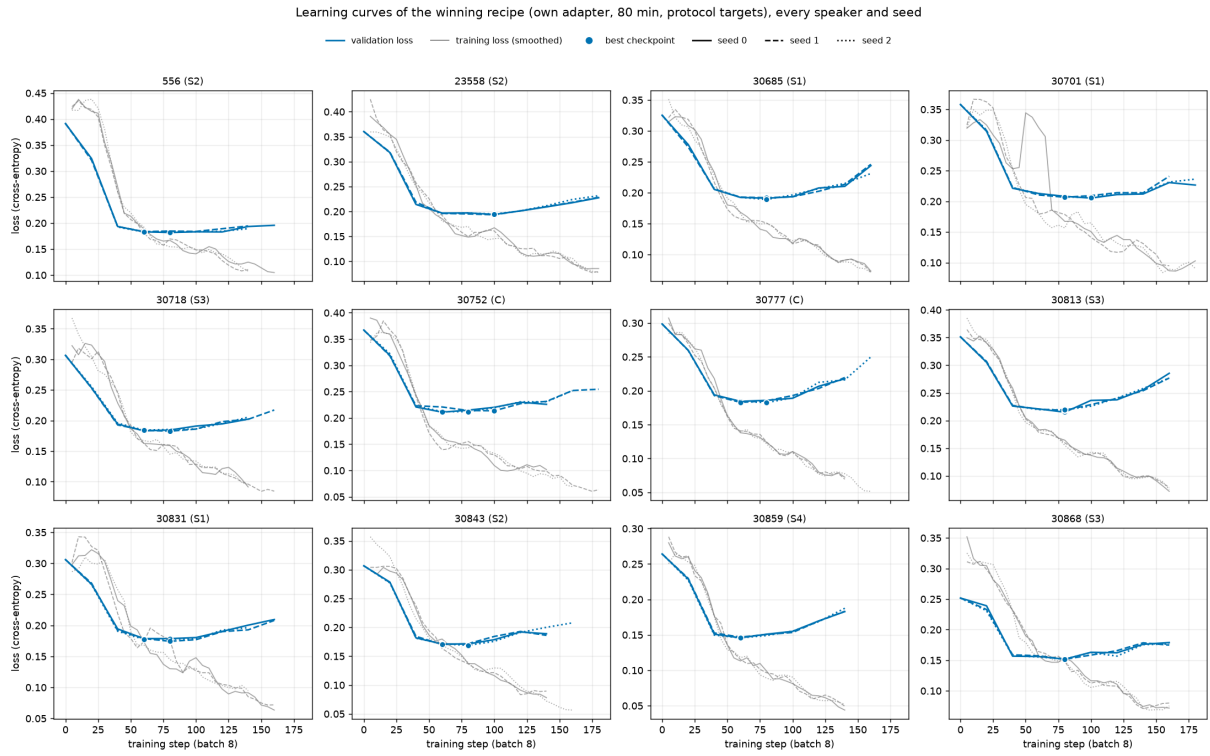


Figure 15: Learning curves of the 80-minute own adapters, one panel per speaker. Blue: validation loss every 20 steps (step 0 is arm B); grey: training loss, smoothed over 20 steps; line style: seed; dot: the checkpoint early stopping restored.